Prompt of CoS2W Experiment

This is the prompt template for the CoS2W experiment and data construction of the SWAB dataset, taking chunk-level meetings as an example.

Prompt for CoS2W

Suppose you are a professional recorder. Please polish the span of meeting in a written and formal style. The refined output should be free of noticeable ASR (Automatic Speech Recognition) errors and grammatical errors, faithfully preserve the original content, and exhibit a clear written style with excellent readability. Pleasure refer to the given meeting abstractive summarization. The given content is a transcript of the meeting using ASR, usually with a spoken or informal style, in the format [[line-id][transcription]. The requirements are as follows:
1. Please correct grammatical, spelling, and ASR errors by referring to the meeting abstractive summarization.
2. Please transfer the informal style such as redundant and repetitive into the formal style.
3. Please improve the fluency and coherence. You can adjust the sentence structure if appropriate.
4. The refined output must faithfully preserve the information of the given content. Do not add, delete or modify important information.
5. Please retain the original language for multilingual content.
6. The number of lines in the polished result must be consistent with the given content. The output format is [line-id][result].
The following is the given meeting abstractive summarization:
{auxiliary}
The following is the given context:
{context}
The following is the given span content of meeting:
{content}
The following is the refined output of the meeting span:

Prompt of Evaluation

We initially attempt to evaluate faithfulness and formality scores within one prompt, but we find that formality scores are highly influenced by faithfulness problems. For samples with faithfulness problems, the LLM Evaluator will award a low score not only in faithfulness but also in formality, even when the formality is satisfied. Consequently, following commonly used methods (Lai, Toral, and Nissim 2023), we independently evaluate the faithfulness and formality scores of the target, which resulted in a good correlation with human evaluations as shown in Figure 3.

Here is the prompt for faithfulness evaluation of the CoS2W task, taking meetings as an example.

Prompt for Faithfulness Evaluation

Suppose you are a professional text editor, please evaluate the quality of the model's refined output. The original content is a paragraph from Automatic Speech Recognition (ASR), which may contain recognition errors, grammatical errors, and spoken/informal style expressions. The model's refined output should be free of noticeable ASR errors and grammatical errors, faithfully preserve the original content, and exhibit a clear written style with excellent readability. The human's refined output can be considered as one of the qualified polished result, which can be used as a reference for quality assessment. Please evaluate and score the model's refined output. Use a scoring range from 1 to 10 for each criterion, where 1 represents the worst performance and 10 represents the best performance. The return fields include:
1. ASR Error Correction Score: Check whether the text effectively corrects errors from automatic speech recognition. The higher the score, the more accurate the correction.
2. Faithful Score: Determine whether the text polishing results maintain the intent and content of the original speech. The higher the score, the more complete the preservation of the original meaning.
3. Evaluation Summary: Based on the scores above, briefly explain the reasons for the scores, provide an overall evaluation of the polishing quality, and suggest modifications.
The following is the original content:
{source}
The following is the human's refined output for reference:
{human-target}
The following is the model's refined output to evaluate:
{model-target}
Please provide scores for each criterion and an overall evaluation with json format:

Given the source and the manually annotated target in the prompt, we use the LLM to evaluate the performance in terms of ASR error correction and faithfulness. However, ASR error correction scores are also significantly impacted by faithfulness problems. The Pearson correlation coefficient (Pearson 1895) between ASR error correction scores and faithfulness scores is as high as 0.96, which indicates nearly perfect correlations. For samples with faithfulness problems, the LLM Evaluator will award a low score not only in faithfulness but also in asr error correction scores, even if there are no ASR errors in the results. Moreover, the correlation between ASR error correction scores and human evaluation is not strong. Therefore, we do not report the ASR error correction scores. Instead, we use the Challenging Keyword Recall (CK-Recall) as the metric for the ASR error correction subtask.

Here is the prompt for formality evaluation of the CoS2W task, taking meetings as an example.

Prompt for Formality Evaluation

Suppose you are a professional text editor, please evaluate the quality of the model's refined output. The model's refined output should be free of grammatical errors, and exhibit a clear written style with excellent readability. Please evaluate and score the model's refined output. Use a scoring range from 1 to 10 for each criterion, where 1 represents the worst performance and 10 represents the best performance. The return fields include:
1. Grammar Score: Evaluate the grammatical accuracy of the text. The higher the score, the fewer grammatical errors there are.
2. Formal Score: Assess the written language style of the text, including the level of formality and suitability for written communication. The higher the score, the stronger the written style of the text.
3. Readability Score: Evaluate the fluency and overall readability of the text, including ease of understanding and natural expression. The higher the score, the better the readability of the text.
4. Evaluation Summary: Based on the scores above, briefly explain the reasons for the scores, provide an overall evaluation of the polishing quality, and suggest modifications.
The following is the model's refined output to evaluate:
{target}
Please provide scores for each criterion and an overall evaluation with json format:

Given only the target result, we use the LLM to evaluate the performance in terms of grammar, formality, and readability. In terms of grammar, we find that nearly all LLM-generated results are free of grammatical errors, as confirmed by both LLM evaluation and human evaluation. Therefore, there is no need to report grammar scores. For readability scores, the Pearson correlation coefficient between readability scores and formality scores is as high as 0.97, which indicates nearly perfect correlations. Thus, the redundant scores have been omitted from the report.

Prompt of Auxiliary Information

Auxiliary information sometimes is comprehensive and detailed with long length, which poses challenges for LLMs in recognizing and utilizing useful information. For example, in the Chinese podcast domain, the average length of auxiliary information is around 2,000 words as shown in Table 9. Therefore, we aim to condense the content to enable LLM to better leverage auxiliary information. Using the LLM, we design two methods: one for generating summaries and another for extracting key entities.

Here are the prompts for extracting summaries and keywords from auxiliary information.

Ranking	Formula	Assigned Coefficient
[1,1,1] vs [1,1,1]	100.0	100.0
[1,1,1] vs [1,1,2]	62.50	50.00
[1,1,1] vs [1,2,2]	62.50	50.00
[1,1,1] vs [1,2,3]	50.00	0.00

Table 8: The Spearman's correlation coefficient for identical rankings. We show both the formula and the assigned coefficient results.

Prompt for Extracting Summaries

Please summarize the provided podcast introduction into a concise English summary. You need to read the following content carefully and write a concise summary. The requirements are as follows:
1. Please accurately understand and recount the key information, ensuring all key points are covered.
2. Preserve all the key vocabulary from the original text, including but not limited to person, location, organization, date, proper nouns, professional terminology, and other named entities, especially those words that automatic speech recognition (ASR) may find challenging to identify correctly.
3. Narrate clearly, with logical coherence, concise language, and avoid redundancy.
The following is the provided podcast introduction:
{auxiliary-information}
The following is the summary:

Prompt for Extracting Keywords

Based on the provided podcast introduction, please accurately identify and extract key entities of different categories. Make sure you read the given content thoroughly and identify and classify keywords from several categories, including but not limited to: person, location, organization, date, proper nouns, professional terms, podcast names, WeChat official account names, etc. Pay special attention to words that automatic speech recognition (ASR) may find challenging to identify correctly. For each category, create a list formatted as follows: the category as the key and the corresponding list of keywords as the value. Please be sure to identify accurately, ensuring the accuracy and completeness of the keywords.
The following is the provided podcast introduction:
{auxiliary-information}
The following is the json result of key entities of different categories:

Spearman's Correlation Coefficient

Table 8 presents the calculation methods of Spearman's correlation coefficient on identical rankings.

Note that different LLMs actually perform quite well in terms of formality. As shown in Table 5, both larger and smaller LLMs achieve a high formality score on the non-blank paragraphs. Thus, annotators generally tend to view

Data	# Words	# Sentences	# Paragraphs	# Turns	# Speakers	# Auxiliary
Chinese	13071.10±7624.45	556.07±297.26	189.07±138.93	169.03±152.45	2.10±1.03	753.63±1059.36
Podcast	21610.10±7878.83	683.10±253.53	202.70±82.16	191.70±94.56	2.90±0.74	2051.30±898.29
Meeting	9926.10±740.61	769.10±151.40	324.80±99.71	314.40±102.57	2.40±0.84	135.70±14.24
Lecture	7677.10±596.89	216.00±25.82	39.70±5.79	1.00±0.00	1.00±0.00	73.90±16.41
English	6039.97±3673.92	426.43±258.67	137.00±94.08	81.73±75.89	2.77±1.74	165.43±74.91
Podcast	10077.40±3605.97	709.00±251.84	232.00±80.39	100.00±51.07	3.30±2.06	192.70±78.33
Meeting	3860.50±1610.35	324.40±111.06	131.60±54.34	144.20±63.24	4.00±0.00	212.60±44.80
Lecture	4182.00±819.03	245.90±57.82	47.40±11.31	1.00±0.00	1.00±0.00	91.00±22.00
SWAB	9555.53±6912.06	491.25±283.89	163.03±120.53	125.38±127.24	2.43±1.45	459.53±801.45

Table 9: The statistic of Spoken2Written of ASR Transcripts Benchmark (SWAB) dataset.

Model	Librispeech test_clean	Gigaspeech test	Wenetspeech net	Wenetspeech meeting
Whisper	**4.00**	**13.50**	8.87	11.25
Paraformer	4.50	14.70	**6.74**	**6.97**

Table 10: The Character Error Rate (CER) on ASR test sets. The smaller metric indicates better ASR recognition performance. Among them, Librispeech and Gigaspeech are the English test sets, while Wenetspeech is the Chinese test set.

LLM outputs as well-written and formal and give them identical rankings (i.e., [1, 1, 1]). The proportion of identical formality rankings is quite high (63.93%) compared to faithfulness ranking (6.06%). However, the identical rankings are not defined in the Spearman correlation coefficient calculations. For distinct rankings $R(X_i)$ and $R(Y_i)$ with n observations, it can be computed using the popular formula:

$$r_s = 1 - \frac{6\sum_{i=1}^{n}(R(X_i) - R(Y_i))^2}{n(n^2-1)}$$

where $R(X_i) - R(Y_i)$ is the difference between the two ranks of each observation.

To ensure these paragraphs are not overlooked, we have assigned the correlation coefficient based on the formula.

Statistic of SWAB

Table 9 presents the statistics of the SWAB. It can be seen that both Chinese and English data contain sufficient paragraphs, providing ample support for the evaluation. Additionally, we have collected relevant auxiliary information. In particular, the podcast domain has a wealth of long text information.

ASR Performance

The paper of the Paraformer (Gao et al. 2022) only reported performance on a Mandarin test set. To illustrate its performance, we present a comparison between the Whisper and Paraformer model on English and Chinese test sets in Table 10. Among them, Librispeech (Panayotov et al. 2015) and Gigaspeech (Chen et al. 2021) are the English test sets, while Wenetspeech (Zhang et al. 2022a) is the Chinese test set. Both models have a similar number of parameters. However, Whisper benefits from over 680,000 hours of training data, while the Paraformer model utilizes only 50,000 hours of training data, giving Whisper a certain advantage in recognition performance.